

# Panel MS-AR(1) on SOE log real GDP per worker

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# 1 Model

Joint panel Markov-switching AR(1) around a trend, with three regimes:

$$y_{it} = \tau_{it} + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it})z_{it} + \sigma(s_{it})\varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

Here  $\tau_{it}$  is a (possibly country-specific) trend in calendar time; the six specifications below differ in how  $\tau_{it}$  is constructed. Countries are independent given shared Markov parameters; latent paths  $s_{it}$  are country-specific. The regime dated  $t$  governs the transition from  $z_t$  to  $z_{t+1}$ .  $\sigma$  switches with the regime. Unless noted,  $\rho$  is regime-specific and the median  $\mu$  is pinned at 0 after ordering. The likelihood is a Hamilton filter per country, summed across the panel.

## Trend assumptions

1. Common intercept  $a$  and common linear slope  $g$  (joint MLE).
2. Country intercepts  $a_i$  and common  $g$  ( $a_i$  profiled).
3. Country intercepts  $a_i$  and slopes  $g_i$  (both profiled).
4. Country-specific Christiano–Fitzgerald low-pass trend (cycle = oscillations of 2–100 quarters / 25 years), then MS-AR on the cycle.
5. Same CF trend, with every  $\mu(s) = 0$ .
6. Same CF trend, with one  $\rho$  common to all regimes.

Sections 4–6 are two-step: the trend is not re-estimated jointly with the cycle. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters only. A dashed entry is a pinned coefficient.

## 2 Common intercept and linear trend

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 17917.39. Common intercept  $a = -4.0009$ . Common slope  $g = 0.0139$ .

Table 1: Regime AR(1) and transition matrix. Columns are regimes.

|          | (0)                | (1)                | (2)                |
|----------|--------------------|--------------------|--------------------|
| $\mu$    | -0.1033            | 0.0000             | 0.7480             |
|          |                    | —                  |                    |
| $\sigma$ | 0.0099<br>(0.0002) | 0.0222             | 0.0692             |
| $\rho$   | 0.9900             | 0.9900             | 0.9900             |
| $\Pi_0.$ | 0.9754<br>(0.0044) | 0.0070             | 0.0177<br>(0.0013) |
| $\Pi_1.$ | 0.0133<br>(0.0023) | 0.9691<br>(0.0029) | 0.0176<br>(0.0021) |
| $\Pi_2.$ | 0.0812<br>(0.0068) | 0.1445<br>(0.0109) | 0.7743<br>(0.0112) |

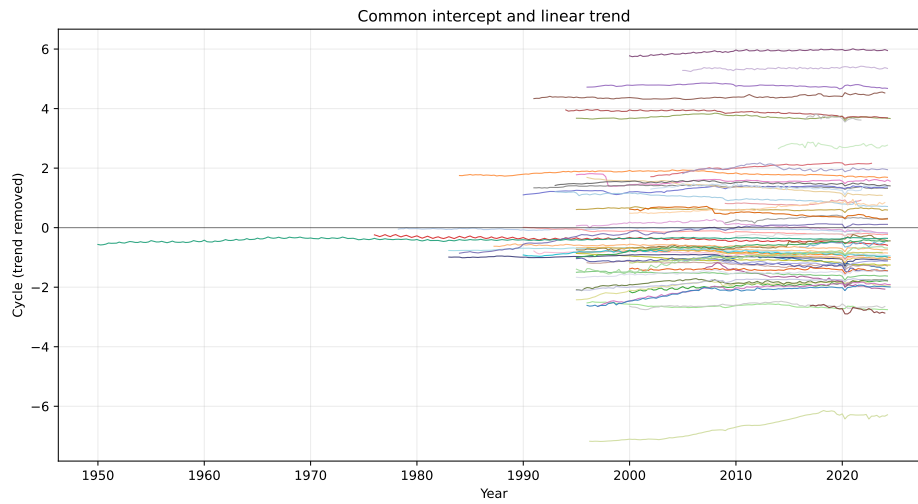


Figure 1: Country cycles after removing the section's trend.

### 3 Country-specific intercepts, common linear trend

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18079.16. Standard errors omitted. Country intercepts  $a_i$ : mean -3.863, min -10.356, max 2.154. Common slope  $g = 0.0115$ .

Table 2: Regime AR(1) and transition matrix. Columns are regimes.

|          | (0)     | (1)    | (2)    |
|----------|---------|--------|--------|
| $\mu$    | -0.1298 | 0.0000 | 0.2260 |
|          |         | —      |        |
| $\sigma$ | 0.0218  | 0.0682 | 0.0097 |
| $\rho$   | 0.9900  | 0.8938 | 0.9892 |
| $\Pi_0$  | 0.9426  | 0.0200 | 0.0374 |
| $\Pi_1$  | 0.0932  | 0.7964 | 0.1104 |
| $\Pi_2$  | 0.0232  | 0.0106 | 0.9662 |

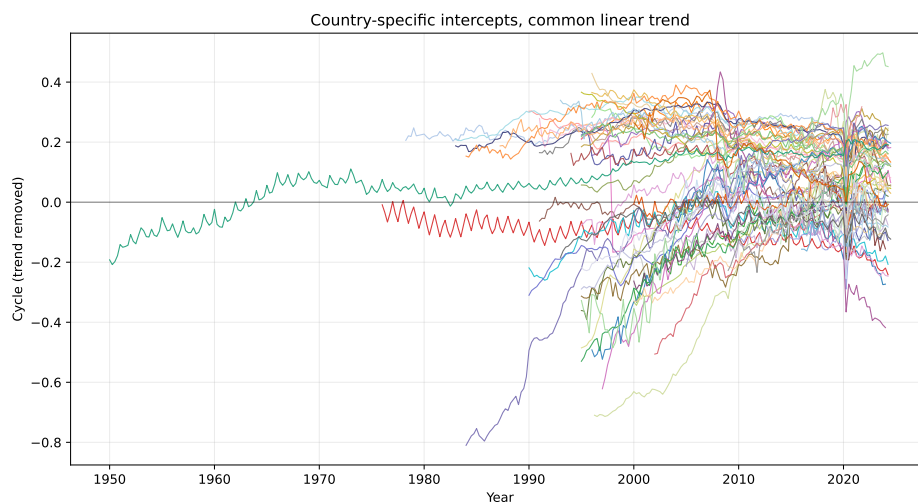


Figure 2: Country cycles after removing the section's trend.

## 4 Country-specific intercepts and linear trends

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18357.24. Country intercepts  $a_i$ : mean -3.974, min -13.117, max 2.031. Country slopes  $g_i$ : mean 0.0140, min -0.0309, max 0.0547.

Table 3: Regime AR(1) and transition matrix. Columns are regimes.

|          | (0)                 | (1)                | (2)                |
|----------|---------------------|--------------------|--------------------|
| $\mu$    | -0.0854<br>(0.0344) | 0.0000<br>—        | 0.0341<br>(0.0005) |
| $\sigma$ | 0.0655              | 0.0094             | 0.0211             |
| $\rho$   | 0.8490<br>(0.0677)  | 0.9751<br>(0.0056) | 0.8981<br>(0.0098) |
| $\Pi_0.$ | 0.7770<br>(0.0263)  | 0.0740             | 0.1491             |
| $\Pi_1.$ | 0.0204<br>(0.0016)  | 0.9717<br>(0.0032) | 0.0080             |
| $\Pi_2.$ | 0.0172<br>(0.0043)  | 0.0179<br>(0.0029) | 0.9648<br>(0.0030) |

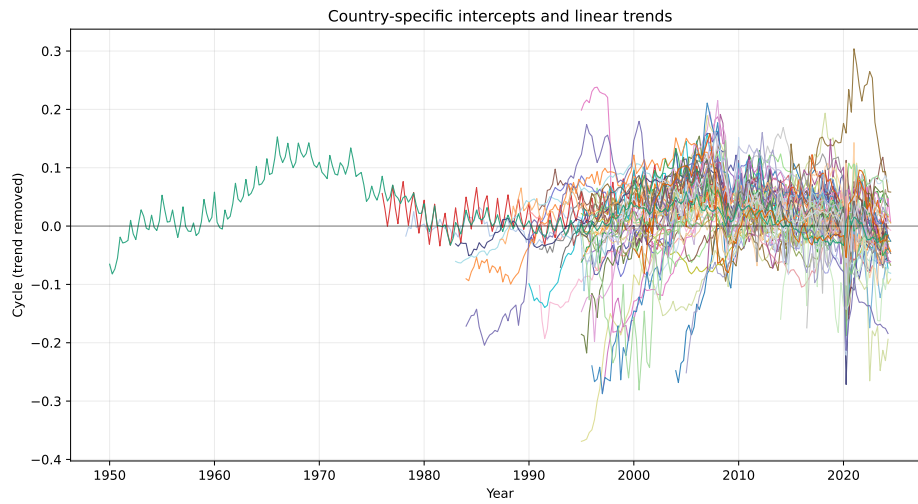


Figure 3: Country cycles after removing the section's trend.

## 5 Country-specific CF filtered trend (25 years)

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18637.48. Christiano–Fitzgerald low-pass trend: periods longer than 100 observations (25 years).

Table 4: Regime AR(1) and transition matrix. Columns are regimes.

|          | (0)                 | (1)                | (2)                |
|----------|---------------------|--------------------|--------------------|
| $\mu$    | -0.0173<br>(0.0054) | 0.0000<br>—        | 0.0027<br>(0.0025) |
| $\sigma$ | 0.0582<br>(0.0026)  | 0.0091<br>(0.0002) | 0.0204<br>(0.0005) |
| $\rho$   | 0.4857<br>(0.0470)  | 0.9242<br>(0.0072) | 0.8484<br>(0.0115) |
| $\Pi_0.$ | 0.7641<br>(0.0317)  | 0.0682<br>(0.0193) | 0.1676<br>(0.0325) |
| $\Pi_1.$ | 0.0191<br>(0.0033)  | 0.9725<br>(0.0034) | 0.0085<br>(0.0031) |
| $\Pi_2.$ | 0.0208<br>(0.0044)  | 0.0164<br>(0.0040) | 0.9628<br>(0.0055) |

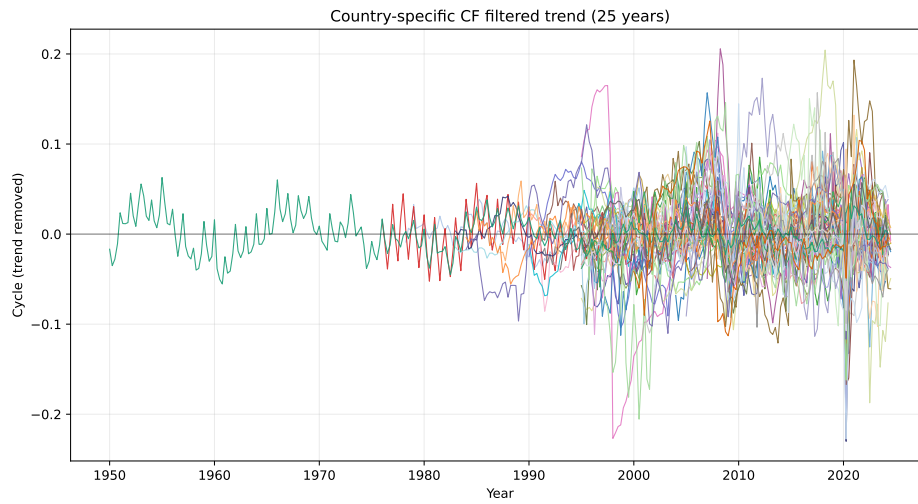


Figure 4: Country cycles after removing the section's trend.

## 6 CF filtered trend, all regime means restricted to 0

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18632.52. Christiano–Fitzgerald low-pass trend: periods longer than 100 observations (25 years).

Table 5: Regime AR(1) and transition matrix. Columns are regimes.

|          | (0)                | (1)                | (2)                |
|----------|--------------------|--------------------|--------------------|
| $\mu$    | 0.0000<br>—        | 0.0000<br>—        | 0.0000<br>—        |
| $\sigma$ | 0.0091<br>(0.0002) | 0.0204<br>(0.0005) | 0.0589<br>(0.0027) |
| $\rho$   | 0.9264<br>(0.0091) | 0.8441<br>(0.0155) | 0.5308<br>(0.0501) |
| $\Pi_0.$ | 0.9723<br>(0.0034) | 0.0082<br>(0.0031) | 0.0194<br>(0.0034) |
| $\Pi_1.$ | 0.0165<br>(0.0040) | 0.9638<br>(0.0055) | 0.0198<br>(0.0044) |
| $\Pi_2.$ | 0.0681<br>(0.0191) | 0.1613<br>(0.0325) | 0.7706<br>(0.0322) |

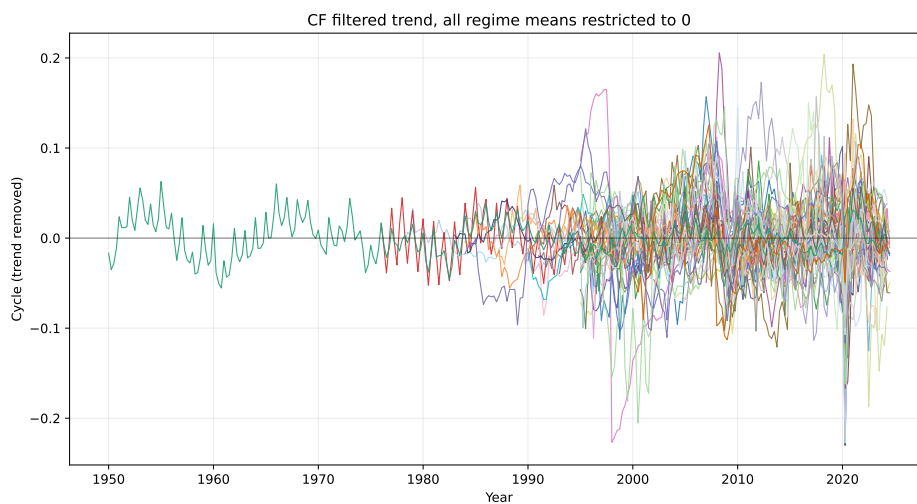


Figure 5: Country cycles after removing the section's trend.

## 7 CF filtered trend, common $\rho$

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18572.70. Christiano–Fitzgerald low-pass trend: periods longer than 100 observations (25 years).

Table 6: Regime AR(1) and transition matrix. Columns are regimes.

|                 | (0)                 | (1)                | (2)                |
|-----------------|---------------------|--------------------|--------------------|
| $\mu$           | -0.0304<br>(0.0246) | 0.0000<br>—        | 0.0017<br>(0.0032) |
| $\sigma$        | 0.0656<br>(0.0030)  | 0.0093<br>(0.0002) | 0.0208<br>(0.0005) |
| $\rho$ (common) | 0.8822<br>(0.0061)  | 0.8822<br>(0.0061) | 0.8822<br>(0.0061) |
| $\Pi_0$ .       | 0.7624<br>(0.0322)  | 0.0661<br>(0.0202) | 0.1715<br>(0.0340) |
| $\Pi_1$ .       | 0.0172<br>(0.0030)  | 0.9742<br>(0.0034) | 0.0086<br>(0.0032) |
| $\Pi_2$ .       | 0.0191<br>(0.0041)  | 0.0166<br>(0.0041) | 0.9643<br>(0.0055) |

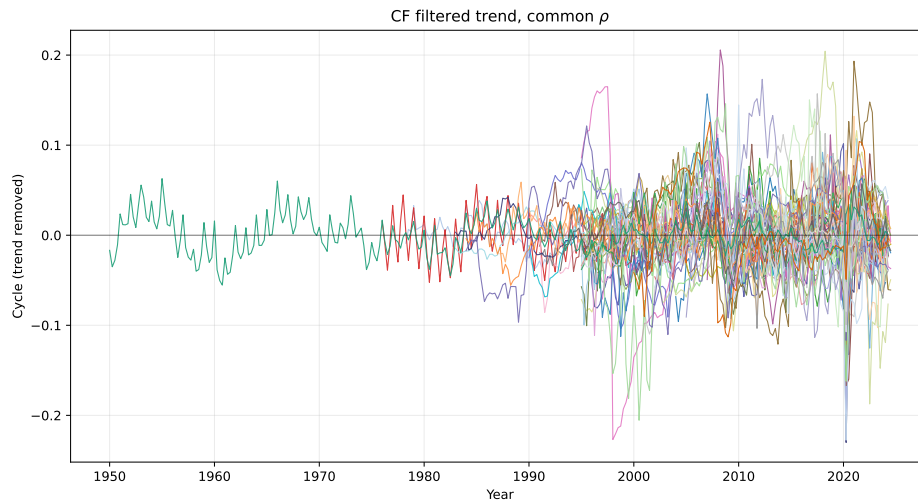


Figure 6: Country cycles after removing the section's trend.